भारतीय राष्ट्रीय भुगतान निगम
NATIONAL PAYMENTS CORPORATION OF INDIA

Embedded Payments

Product Document

Version 1.3

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Glossary

1) PSP_e - PSP on the embedded device i.e., car/tv/speaker etc.
2) PSP_0- PSP on the primary device i.e., mobile phone
3) Primary App – UPI App on the primary device
4) Embedded App - Application on the embedded device

1) Background

Embedded/ IOT (Internet of Things) based payments refer to the payments that can be made via various devices to eliminate the need for dependency on single device to make UPI payments in a secure manner, providing a seamless experience to the users.

In line with the vision of Reserve Bank of India, NPCI aims to bring revolutionary technology to provide the best payment experience to the people of India. In India, currently the payment methods and infrastructure depend mostly on mobile phone.

2) Scope

Embedded/ IOT Payments in India reduces reliance on physical cash and dependency on single device to make payments. Aligning with RBI's vision, NPCI aims to bring these payment methods for the people of India ensuring the security of transactions and data privacy.

The scope of this will include but not limited to:

1. Connected Cars – Embedded/ IOT in automobile will enable UPI Payments for tolls, parking, fuelling, maintenance etc. reducing the need for a single device. Users will be able to make the payment through the infotainment system of the car without the need for wallets/cash/mobile phones

2. Smart Home Appliances – Appliances like Televisions, refrigerators, voice assistant devices will be able to make payments for subscriptions, groceries, bill payments respectively

3. Wearable Devices – Smartwatches, fitness trackers, and other wearables can be equipped with UPI Payment capabilities allowing users to make payments on the go. Payments will be made with just a tap on the POS devices or the smart QRs

Apart from these use-cases other industries where this can be implemented down the line will include transportation, hospitality, healthcare etc.

3) Benefits of Embedded Payments to Consumers and Merchants

Consumers

Benefits to the consumers include the following:

1) Convenience – Allows convenient and seamless payments without the need for physical cards or cash. Consumers can pay directly from their smart watches, connected cars, or other embedded devices without the need for mobile phones
2) Speed – Consumers can discover merchants and make payments faster as compared to traditional payment methods reducing waiting times
3) Accessibility – Can be accessed from various devices and locations providing consumers with greater flexibility and accessibility
4) Security – Embedded Solution will make sure that the payments made through embedded devices will be as secure as the existing UPI mechanism
5) Integration – Embedded Payments can be seamlessly integrated into consumers' everyday lives making it easier to manage finances, automate recurring payments etc.

Merchants

Benefits to the merchants include the following:

1) Streamlined Transactions – Embedded Payments will result in faster and efficient payments and hence lead to shorter checkout times and improved customer satisfaction
2) Enhanced Customer Experience – Merchants who would enable acceptance of embedded payments will lead to higher customer satisfaction and loyalty since it will give a seamless payment experience to the customer
3) Increased Sales – These payment methods will attract more customers and increase sales
4) Data Insights – Merchants will get insights on customer behaviour and hence can make use of personalized marketing efforts, improve inventory management, and optimize pricing strategies

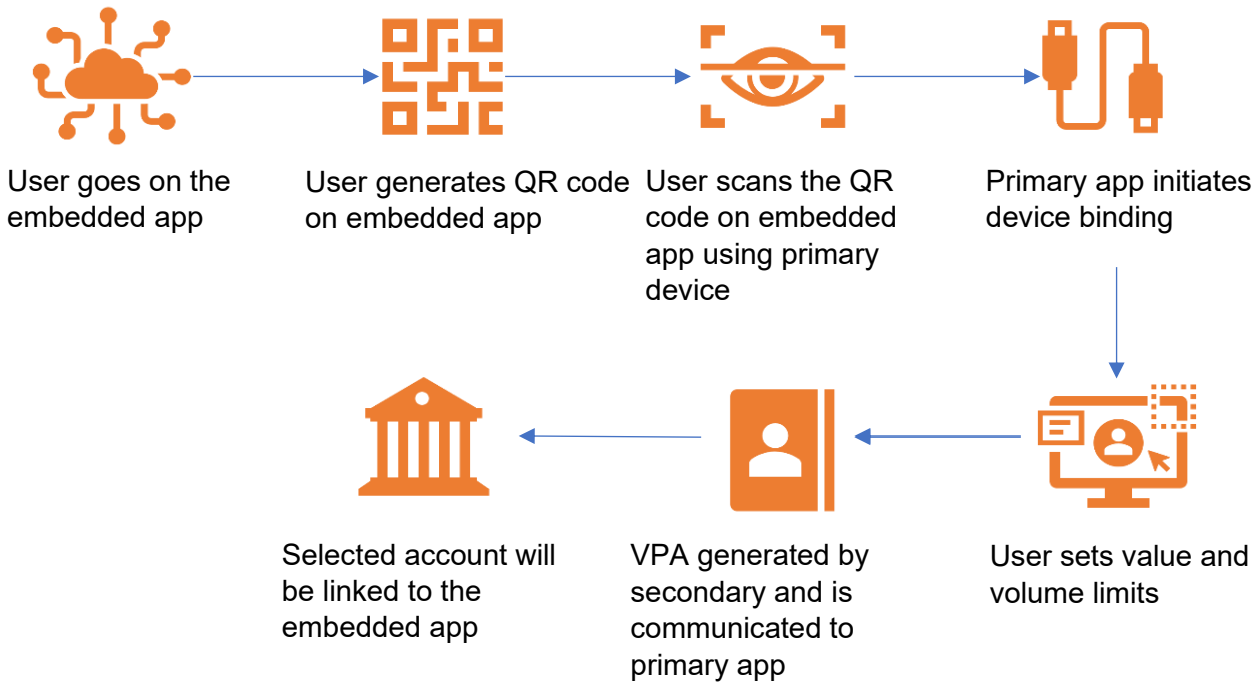
4. Proposed Solution

This solution which is app based for Embedded Payments aims to bring device binding and CL into embedded device. The app will be used on primary device and on embedded device. Device binding and registration for the embedded app will be done through the primary device and the payment will be initiated through the embedded app. For security purposes, the limit setting and deregistration of the embedded app will be allowed through both the primary app on the primary device and the secondary app on secondary device.

STEPS

1) User signs up on the primary mobile app (primary device), after signing up the user scans the QR code from the embedded app to populate embedded app details like device ID/VIN, app ID, mobile number, org ID, ref ID, device type, manufacturer etc.
2) User starts device binding and registration of embedded app from primary device
3) Account is selected for the registration of embedded app which was set for the primary device
4) VPA is generated for the Embedded app
5) VPA is communicated to the Primary app

Note – All the infosec protocols applicable to UPI payment app will be followed by the embedded payment app provider and payment service provider



User Registration journey

1) User will access the embedded app and for the first time will initiate device binding and registration by generating a QR code on the embedded app. The QR code will have device details like device ID/VIN, app ID, mobile number, org ID, ref ID, device type, manufacturer etc
2) User will have to use the mobile for the first time, get onboarded on the primary app and then scan the QR generated on the secondary device using the app provided by the same or different app provider to the embedded app
3) Once the device details have been captured on the app in the primary device, device binding is initiated where the app provider communicates the device details, mobile number, channel, org ID, app ID, ref ID to the PSP & NPCI
4) These details are sent to PSPo as well as PSPe via NPCI. PSPo generates random number and sends VMN to primary app for device binding
5) Default limits for the embedded app like the transaction limit (value), expiry will be assigned to the dependent app, which can be changed after the cooling period
6) Account details request is raised by PSPe, this is sent to NPCI, NPCI sends to PSP_0 and PSPe generates the VPA using the account details sent by PSP_0 via NPCI. This is then communicated to both dependent and primary app. [APIs used are RegED (SetUPI ID) and List Account as stated in TSD]
7) VPA is communicated to the embedded app by PSP_e i.e. for e.g., type. alphanumeric@handle. Value of 'type" will be taken from the QR generated initially on the embedded device. Type is defined in the QR specification
8) A purpose code is defined for embedded payments transactions
9) Channels and modes as per the table below

Type	Mode (Payment)	Mode (Autopay)	Mode (OTM)	Channel
Car	33	34	35	EMDCC
Speaker	33	34	35	EMDSPK
Fridge	33	34	35	EMDFR
Wearables	33	34	35	EMDWR
TV	33	34	35	EMDTV
Other1	33	34	35	TBD""
Other2	33	34	35	TBD""
Other3	33	34	35	TBD""
Other4	33	34	35	TBD""

TBD"" – For future use cases

10) Account details from the primary app will be sent to PSPe by PSP_0 during account detail request
11) User will have an option to deregister/logout from the embedded app or the primary app
12) Provision for cooling period post registration. During cooling period, a limit on the transaction will be kept. (Limits will be defined)

5. Roles and Responsibilities

Primary App	Embedded app	PSPo Primary PSP	PSPe Embedded PSP	NPCI	OEM
To provide an interface to the user for adding embedded device	To provide an interface to make payments from embedded device with UPI CL	Communicate with PSPe via NPCI	Store limits for the embedded app	Route details between the PSPs	Passing of required details for QR creation
Enable SMS binding for embedded device	To facilitate payments from embedded device	Allow SMS binding for embedded app	Generate VPA for embedded app	Ensure PSP that info is from a trusted source	Timely app updates to be facilitated
Allow limit setting for embedded device		Transfer the limits to PSPe	To facilitate payments from embedded device		
Allow deregistration for embedded device		Provide account details to PSPe for VPA generation	Responsible for payment security		
		Keep master control of embedded device	Communicate with PSPo on regular basis		

Primary App

1) Primary App should have an option to scan embedded app QR and initiate registration
2) Primary App should take **consent** from the user before SMS binding is initiated for the embedded app
3) The Primary App should be able to do **initiate device binding** with SMS consent for the embedded device getting added by the user
4) Primary App should provide option to **select account number** from the list of accounts that are already added to the app
5) Post SMS binding and list account, the Primary App should give an option to set limits for the embedded app to the user
6) The Primary App should have an option for the user to **check the balance limits** for the embedded device
7) Primary App should provide an option to **deregister the embedded device**
8) For any future modifications in the limits by user via Primary App, it must ensure that the particular **account is STILL LINKED in Primary App**. In case the account is not linked, Primary App should prompt user either to link the account again or deregister the embedded device
9) PRIMARY APP to have a **section to display the added embedded devices**
10) Whenever a new device is added in PRIMARY APP, it should provide a **free text to add a nickname** to the added device by user for ease of identification. for e.g., the user can give an identifier to the VPA as BMW car, in order to identify that the VPA belongs to that particular car
11) Primary App should have dispute raising mechanism through UDIR for the transaction done through secondary device

Embedded App

1) The App should generate a QR code to be scanned by PRIMARY APP for SMS binding
2) App will connect with Primary App to receive RefID and OrgID before generation of QR
3) The App should facilitate payments through embedded device
4) Embedded app should have an option to deregister the embedded device
5) Embedded app should have a section for the user to check the available limits
6) Embedded app will have to register itself with the CL to complete device binding
7) The app should have the information of the type of device it is embedded in. For e.g., car, refrigerator, TV etc.
8) PIN cannot be set through Embedded app

9) Limits cannot be set through this app for embedded app
10) App providers for embedded device should have dispute raising mechanism through UDIR
11) All devices should have a unique device ID which needs to be shared during device binding to PSPo.

PSP_0

1) It should facilitate SMS binding for embedded device
2) It should facilitate limit set up for embedded device
3) Deregistration should be facilitated for embedded device
4) PSP_0 should forward the account details linked to PRIMARY APP initially to PSP_e for registration and VPA generation
5) New initiation modes and initiating channels need to be added
6) UDIR functionality to be provided for the transactions done through embedded device. The complaints to be raised with secondary PSP details and secondary PSP VPA by the primary PSP to make the back tracking of transactions seamless

PSP_e

1) It should facilitate merchant payments through embedded device
2) Embedded device limits to be stored by PSP_e and regular update of the limits to be take care via inter PSP communication
3) PSP_e should make sure that the cooling period for the embedded app is adhered to
4) It should provide necessary information to Primary App to generate QR for the PRIMARY APP to scan
5) New initiation modes, initiating channels and purpose code needs to be added
6) It should generate and link VPA based on account information received from PSP_0
7) Deregistration to be facilitated
8) It should generate a reference ID in the start during QR code generation for security
9) One secondary device cannot have multiple primary PSPs
10) Query tag generation in QR should be in “DeviceID | AppID | MobileNumber | Initiating Channel | OEM Details” format

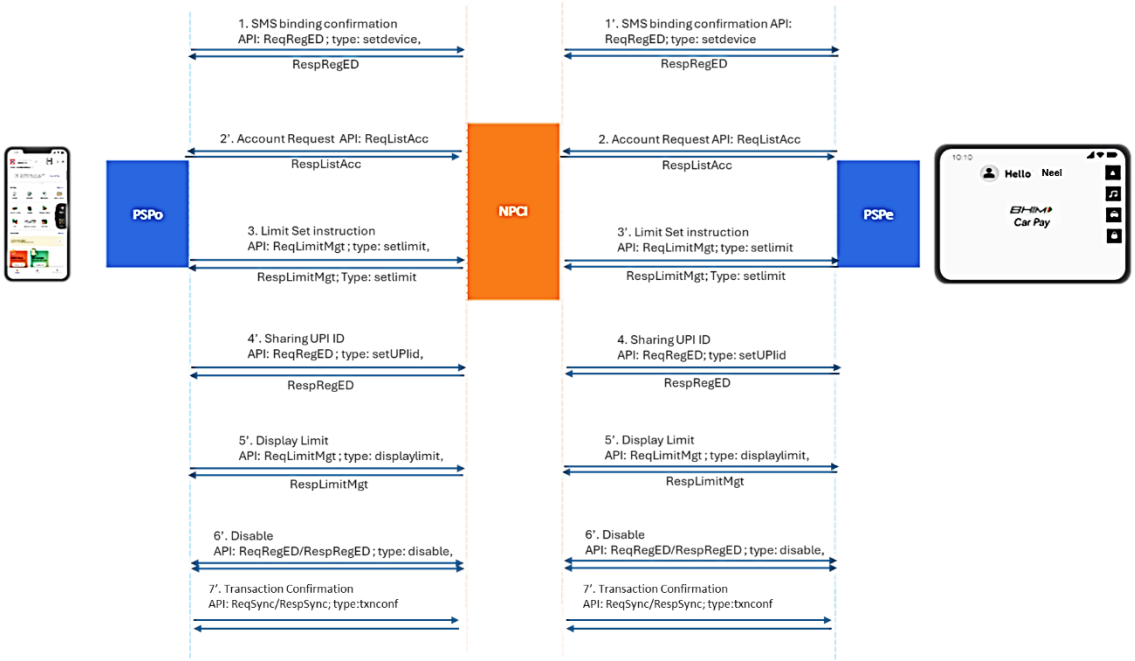
OEM

1) OEM should pass the device id, type of device and mobile number used during device registration to secondary app to generate the QR
2) OEM should provide regular updates for the embedded application on embedded device

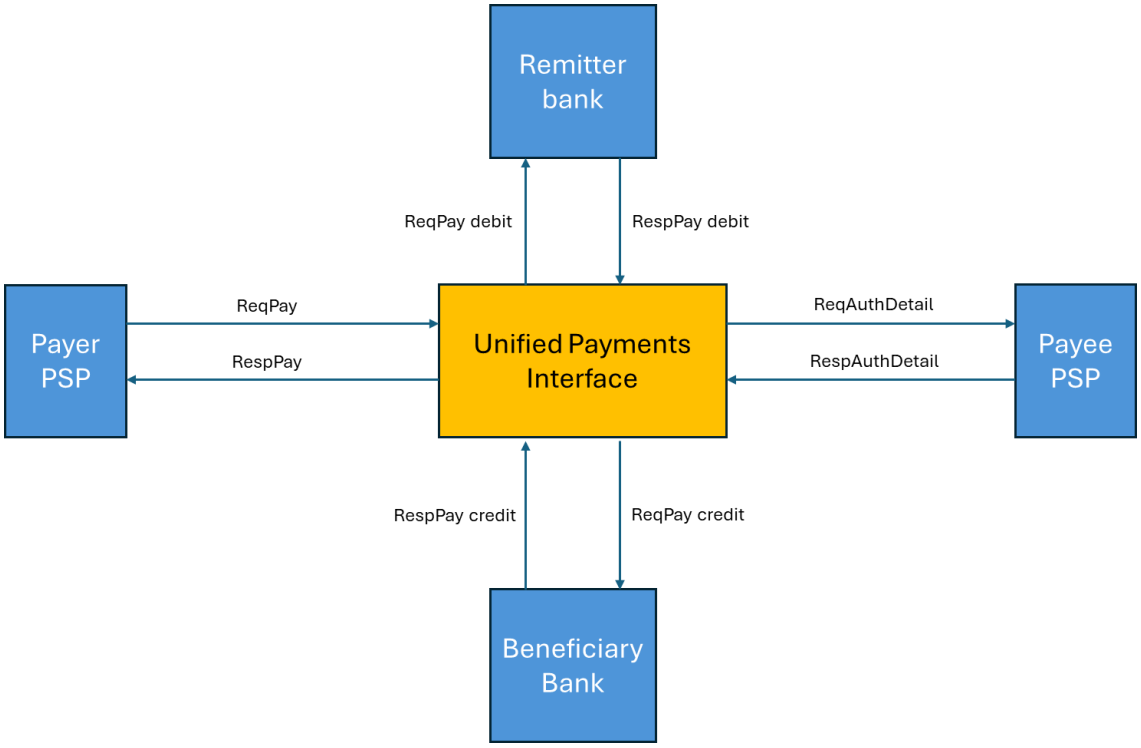
6. Risk and Compliance

1) Cooling period set for Embedded/ IOT devices for first 24 hours
2) Value limits for the embedded device can only be set through the primary app
3) User consent is taken on PRIMARY APP before device binding is initiated
4) Deregistration allowed from the primary app for the embedded app
5) Deregistration allowed from the embedded app as well
6) User can change or replenish the limits whenever through the primary device
7) UPI app on mobile i.e., PRIMARY APP will be only responsible for device binding i.e., SMS verification confirmation, post which Primary App will be responsible for CL registration with NPCI
8) Limit instructions will be sent from primary device but validation and value storage will be done by PSP_e
9) While creation of QR PSP_e, a session reference number which needs to be passed in the QR for registration. This reference number can be valid for 10 minutes, post which any request with the same reference number should be rejected. Also, PSP_e should check the reference number in case of any request

7. List of APIs



Payment flow in embedded device



Description :

1) ReqRegED, RespRegED, ReqLimitMgt, RespLimitMgt, ReqSync, RespSync – New API which includes TYPES : SetDevice, SetLimit, SetUPIid, Displaylimit, Disable, TXNCONF.

   SetDevice – To give SMS binding confirmation to PSPe from PSP0 via NPCI
   SetLimit – To set limits value, volume and expiry for Embedded App from PRIMARY APP
   SetUPIid – To send the new VPA generated for Embedded App to NPCI and PSP0
   Display Limit – To fetch the pending limit details of Embedded App from PSPe. Initiated from PRIMARY APP
   Disable – To disable UPI on embedded device
   TXNCONF – To share transaction details to PSP0 from PSPe

2) ListAccount – This is the existing API which will be modified to bring the account details to PSPe from PRIMARY APP/PSP0 so that VPA can be generated based on any of the accounts linked to PRIMARY APP

3) Other than these APIs, ReqPay, ReqAuthDetail will be as per existing flow.

8. Settlement and Reconciliation

- Settlement of UPI Credit transactions and dispute management will be as per the UPI settlement process.
- NPCI will carryout settlement pertaining to embedded payments in 8 payment cycles and transactions will be included in existing Raw file
- Embedded payment transactions will have dedicated identifiers as purpose code in the back office
- The file will contain transactions of embedded UPI payment usage in the given cycle